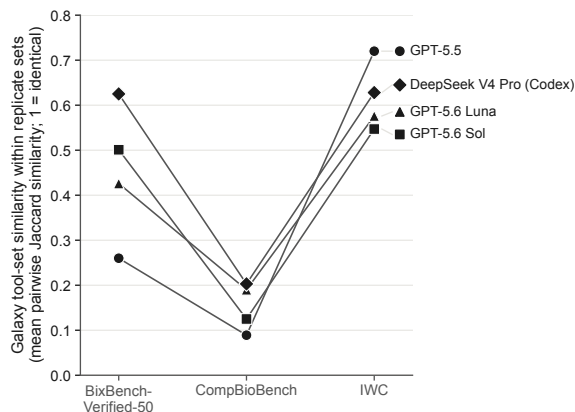


a Open-ended code condition runs spread across more command-line software than Galaxy-condition runs

	Open-ended code	Galaxy	Open-ended code	Galaxy	Open-ended code	Galaxy
	BixBench-Verified-50		CompBioBench		IWC	
phykit	12	14	-	-	-	-
DESeq2 or PyDESeq2	17	7	-	-	20	11
gseapy	8	6	-	1	-	-
samtools	4	2	28	10	40	12
bctools	2	2	12	5	8	-
bedtools	-	-	14	11	13	2
anndata	2	-	10	7	10	7
scanpy	1	-	9	4	10	-
datamash	-	8	-	9	1	-
bwa	2	2	14	6	26	4
fastp	-	-	5	-	28	20
bowtie2	-	-	14	3	22	20
minimap2	1	-	19	2	27	1
MACS2 or MACS3	-	-	5	1	10	8
edgeR	4	-	-	-	14	10
dada2	-	-	-	-	10	9

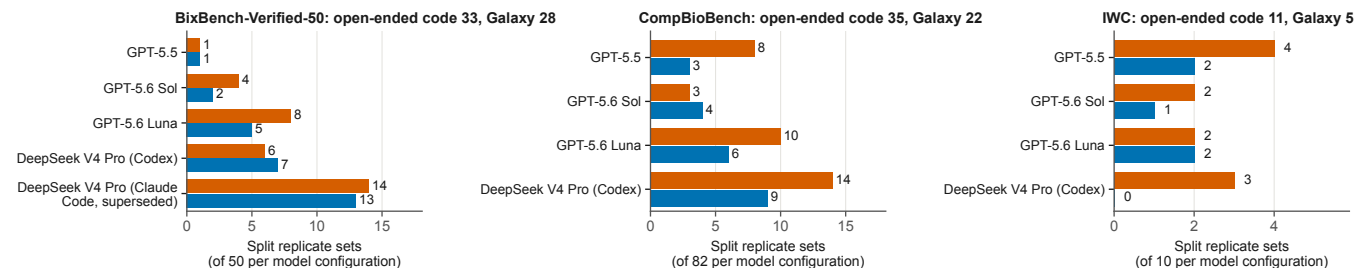
Cell value: % of runs using the software (three GPT model configurations shared by all benchmarks).
Open-ended code condition: command names in shell commands. Galaxy condition: installed Galaxy tools in job records.

b Galaxy tool-set similarity is highest on the workflow-derived benchmark



c The Galaxy condition had fewer split replicate sets than the open-ended code condition in every benchmark

A replicate set is the three replicate runs of one task × model configuration × execution condition. It is split when its runs disagree (BixBench-Verified-50: 1 or 2 of 3 scored correct; CompBioBench: 1 or 2 of 3 matched the consensus answer given by at least 20 of 25 runs, 82 tasks; IWC: output agreement ranged by more than 0.05 within the set).



d Replicate runs diverged for different reasons in each execution condition (BixBench-Verified-50)

Divergence mechanism: a hand-written method or software-version difference in 67% of open-ended code condition runs and 14% of Galaxy-condition runs; a Galaxy interface trap in 17% of Galaxy-condition runs and

